

Arunabha Majumder

Mechanical Design & Robotics R&D Engineer — Actuators, Controls & Hardware Systems

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SUMMARY

Mechanical design engineer who takes electromechanical systems from concept to tested hardware. Designed an actuated exoskeleton that cut energy use by more than 25% (Best Paper Award, 2026) and developed a force-estimation method accurate to below 10% error. Hands-on with SolidWorks, FEA, prototyping, sensors and control.

KEY RESULTS

- **>25%** energy reduction from a hybrid actuation design (Best Paper Award, IFToMM ISRM 2026)
- **<10%** RMS error on sensorless force estimation, validated against a force sensor
- **0.3–0.78%** tracking error on a pneumatic actuator with a neural-network-based controller

EXPERIENCE

Research Engineer – Mechanical Design & Actuator Development (PhD Project: VIEXO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Cut energy consumption by more than 25% in prototype testing of an actuated shoulder exoskeleton by taking technical ownership of the concept, mechanical architecture and design decisions for a hybrid actuation system that pairs a parallel spring with a 6 Nm motor (Best Paper Award, IFToMM ISRM 2026).
- Delivered a working prototype from concept to tested hardware by designing the structure and assemblies in SolidWorks, using structural FEA to balance strength, stiffness and weight, choosing between machined and 3D-printed parts for manufacturability, and integrating the motor, encoder, force sensors and DAQ.
- Achieved force estimation accurate to below 10% RMS error, validated against a force sensor, by developing a sensorless method for a 5-bar planar parallel robot using variable-stiffness actuators and the robot Jacobian; applied the method to measure human arm impedance in the gait lab.
- Reduced user muscle effort by more than 15%, measured with EMG in tests with participants, by implementing an Assist-as-Needed controller on the prototype.
- Collected a full EMG and motion-capture dataset from a 12-participant study across 3 assistance modes and 2 load conditions, by running the data collection within a multidisciplinary team spanning mechanical, electronics, control and biomechanics.
- Mentored and supervised Bachelor's (B.Tech) students on engineering projects (2023 – 2025) by guiding their design work, and taught university students.

MTech Researcher – Pneumatic Actuator Systems (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Reduced position-tracking error to 0.3–0.78%, outperforming classical PID under varying load, by designing a neural-network-based gain-scheduled PID controller for a non-linear pneumatic actuator.
- Built an empirical actuator model from test data by fabricating pneumatic artificial muscles in-house and setting up a test rig with valves, pressure sensors and DAQ.

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

AWARDS & CERTIFICATION

- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- Certified SolidWorks Associate (CSWA) – Mechanical Design
- TATA Technologies Innovation Grant – INR 100,000 (2019); Best Major Project Award (2019)

SKILLS

- **Mechanical Design:** SolidWorks (CSWA), Fusion 360, structural FEA, assemblies, technical drawings, DFA, manufacturability
- **Prototyping & Testing:** Machining, 3D printing, assembly, test rigs, DAQ, data collection
- **Mechatronics & Control:** Motors, encoders, force sensors, pneumatics, embedded control, MATLAB/Simulink, Python
- **Instrumentation:** Qualisys motion capture, EMG, Forsentec force/torque sensors, LabVIEW (basic), Arduino

PUBLICATIONS

- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer.
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.

ADDITIONAL

- **Hands-on builds:** Rehabilitation exoskeleton (Best Major Project Award, 2019; video: youtube.com/watch?v=q5Ystz9NCpQ), Go-Kart (2017), Efficycle human-electric hybrid vehicle (2018)
- **Visual skills:** Diploma in Fine Arts (Painting), Bangiya Sangeet Parishad; used for technical sketching and illustration
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

REFERENCES

Available upon request.